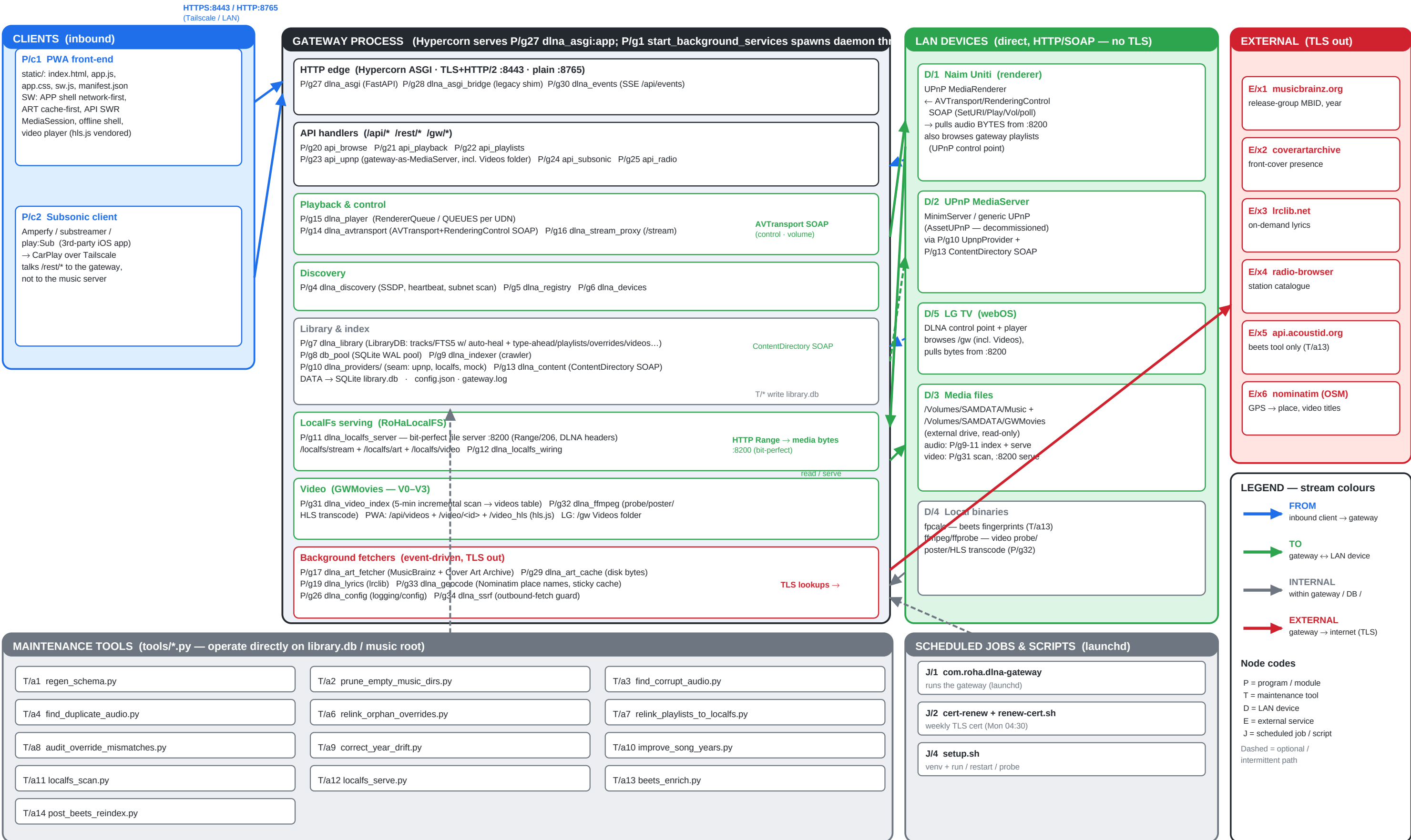


## DLNA Gateway — Architecture (2.1 · ASGI/Hypercorn)

What runs where · stream colours show direction/scope · node codes (P/T/D/E/J) index into the lists overlaid



Node lists — every box in the drawing

Codes match the diagram. **P** = program / module, **T** = tool, **D** = device, **E** = external service, **J** = scheduled job.

Application programs & modules (P/\*)

| Code  | File / location                                          | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P/c1  | static/index.html, app.js, app.css, sw.js, manifest.json | PWA front-end. Browse/playback UI, MediaSession lock-screen, Service-Worker caches (APP shell network-first since 2026-06-27 — cache is the offline fallback only, ART cache-first, API stale-while-revalidate), offline shell. Videos section with a same-origin <video> player + vendored hls.js for the on-demand HLS transcode.                                                                                                                                                                                                                                                                                                                                               |
| P/c2  | (3rd-party iOS app — Amperfy / substreamer / play:Sub)   | Subsonic client for CarPlay over Tailscale. Talks /rest/* to the gateway; never to the music server directly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P/g1  | dlna_gateway.py                                          | Module-wiring + start_background_services(): spawns the daemon threads (SSDP listener, pre-prober, subnet scanner, heartbeat, gateway announcer). Called from the dlna_asgi lifespan so `hypercorn dlna_asgi:app` boots the whole gateway. Its own stdlib HTTP edge + TLS were removed in 2.0 (Hypercorn owns the edge).                                                                                                                                                                                                                                                                                                                                                          |
| P/g3  | dlna_routes.py                                           | GET_ROUTES / POST_ROUTES path→handler maps; the ASGI bridge (P/g28) mounts the not-yet-native ones.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P/g4  | dlna_discovery.py                                        | SSDP multicast listener, probe, subnet scanner, renderer/server heartbeat. Holds SERVERS / RENDERERS singletons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| P/g5  | dlna_registry.py                                         | Data classes + thread-safe ServerRegistry / RendererRegistry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| P/g6  | dlna_devices.py                                          | DeviceRoleCache — in-memory mirror of device_roles for zero-latency classification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P/g7  | dlna_library.py                                          | LibraryDB — SQLite index, FTS5 search, playlists, album_art, play_counts, lyrics, metadata_overrides, favourites, radio, videos + geocode_cache. Composition root for DB-owning singletons + fetchers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P/g8  | db_pool.py                                               | SQLite connection pool — WAL, thread-local conns, write serialization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P/g9  | dlna_indexer.py                                          | Background crawler that walks a provider and populates LibraryDB (with FTS self-heal).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P/g10 | dlna_providers/ (__init__, upnp.py, localfs.py, mock.py) | LibraryProvider seam + registry. UpnpProvider wraps the UPnP SOAP path (kept for MinimServer); LocalFsProvider is the in-process backend (mutagen, content-hashed ids, watchdog).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| P/g11 | dlna_localfs_server.py                                   | RoHaLocalFS bit-perfect file server (:8200). Range-aware (206/416), DLNA-headered, /localfs/stream/ + /localfs/art/ + /localfs/video/ (the LG plays videos from here). Path-traversal defended.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P/g12 | dlna_localfs_wiring.py                                   | Boot-time wiring of the LocalFs provider (maybe_start_localfs); gated on LOCALFS_MUSIC_ROOT / config. Also starts the GWMovies video scan (P/g31, VIDEO_UDN) when LOCALFS_VIDEO_ROOT / localfs.video_root is set — incremental, re-run every 5 min.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P/g13 | dlna_content.py                                          | UPnP ContentDirectory SOAP client (cd_browse/cd_search). Reached only via P/g10 upnp.py.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P/g14 | dlna_avtransport.py                                      | UPnP AVTransport + RenderingControl SOAP (SetURI/Play/Stop/Pause, state/position probe, SetVolume).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| P/g15 | dlna_player.py                                           | RendererQueue (sequential per-renderer playback, gapless via SetNextAVTransportURI, watchdog) + QueueRegistry (QUEUES, one per UDN).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P/g16 | dlna_stream_proxy.py                                     | Browser-audio HTTP proxy /stream (Range pass-through, 5-min idle timeout) + proxy_radio_stream /radio_stream (ICY de-interleave).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| P/g17 | dlna_art_fetcher.py                                      | AlbumArtFetcher — MusicBrainz release-group + Cover Art Archive lookup; sticky notfound cache; ~1 req/s.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P/g19 | dlna_lyrics.py                                           | On-demand IrcLib lyrics; cached in the lyrics table; sticky positive + negative.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| P/g20 | api_browse.py                                            | Browse / search / radio-shuffle API.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P/g21 | api_playback.py                                          | Playback, /stream route, /art proxy, /api/client_log, state, indexer management.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| P/g22 | api_playlists.py                                         | Playlist CRUD endpoints.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| P/g23 | api_upnp.py                                              | The gateway-as-MediaServer: a COMPLETE DLNA Media Server the Naim/LG browse. device.xml (MediaServer + X_DLNA DOC + icons + ContentDirectory AND ConnectionManager), both SCPDs, ContentDirectory#Browse over the full library + the pre-browse handshake actions, ConnectionManager #GetProtocolInfo, GENA SUBSCRIBE + initial NOTIFY, SSDP announce + M-SEARCH responder. Root also lists a Videos folder (only when videos exist) so the LG TV browses + plays GWMovies natively (HEVC/MKV incl.).                                                                                                                                                                             |
| P/g24 | api_subsonic.py                                          | Subsonic-compatible /rest/* API (browse, playlists, favourites→starred, stream, cover, scrobble, internet radio) for CarPlay clients.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| P/g25 | api_radio.py                                             | Internet-radio endpoints: radio-browser search, favourites (≤25), ICY now-playing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| P/g26 | dlna_config.py                                           | Constants (DB_FILE/CFG_FILE/LOG_FILE), logging setup, config load/save, .env load (if dotenv present), raise_fd_limit(8192).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| P/g27 | dlna_asgi.py                                             | 2.0 — THE server (Hypercorn owns the whole edge). FastAPI app: TLS+HTTP/2 on :8443 via ALPN, plain :8765; owns the tailscale cert. Native routes for the read API, /art, /stream + /radio_stream relays, static/PWA, the Subsonic byte methods, the video surface (/api/videos, /video/<id>, /video_poster, /video_transcode, /video_hls — same-origin so the PWA's <video> works over HTTPS), and the Naim/LG-facing /gw/* UPnP surface (device.xml/desc.xml/events/control on the plain :8765 bind — Cleanup C folded it in here, retiring the separate device server). Remaining legacy handlers run via the bridge. Lifespan boots P/g1 services. docs_url off (no CDN call). |
| P/g28 | dlna_asgi_bridge.py                                      | Shim that runs the legacy (h, params) handlers unchanged inside the ASGI app (a fake `h` captures _json/_html/_xml/send_error; runs in a threadpool). Routes are rewritten native one batch at a time, then dropped from the bridge.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| P/g29 | dlna_art_cache.py                                        | On-disk cover-art byte cache keyed by source URL. art_fetch_cached() (in P/g21) fronts art_fetch so /art and Subsonic getCoverArt serve repeat covers from disk (across clients + restarts) instead of re-fetching coverartarchive / re-decoding embedded art. TTL + size capped; art_cache/ gitignored.                                                                                                                                                                                                                                                                                                                                                                          |
| P/g30 | dlna_events.py                                           | EventBus/EVENTS — thread-safe publish to the asyncio loop; native GET /api/events (SSE). Publishers: RendererQueue state, index status transitions, discovery changes. The PWA opens an EventSource as a polling accelerator (fallback intact).                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| P/g31 | dlna_video_index.py                                      | GWMovies scanner (V1) → the videos table. Stable id = sha1(rel_path); metadata via P/g32 probe (filename/mtime fallback when ffprobe absent); place name via P/g33; poster frame extracted per video. Incremental (mtime,size) + prunes gone files; force=True rebuilds.                                                                                                                                                                                                                                                                                                                                                                                                          |
| P/g32 | dlna_ffmpeg.py                                           | Optional ffmpeg/ffprobe helpers (V0/V3, same posture as fpcalc — absent binaries degrade gracefully). probe() metadata/GPS/capture time, extract_poster(), transcode_cmd() for the on-demand H.264/AAC HLS transcode (-pix_fmt yuv420p so 10-bit HEVC plays), build_display_title() for the '<place>_YYYYMMDD_HHMM' fallback title.                                                                                                                                                                                                                                                                                                                                               |
| P/g33 | dlna_geocode.py                                          | Reverse-geocode GPS → place name via Nominatim/OSM (E/x6), cache-first in geocode_cache. UA + 1.1 s rate limit per OSM policy; failures NOT cached (retry later), definitive no-name cached sticky as ". PRIVACY: this sends your clips' GPS coordinates off-machine automatically; opt out by not setting LOCALFS_VIDEO_ROOT.                                                                                                                                                                                                                                                                                                                                                    |
| P/g34 | dlna_ssrf.py                                             | SECURITY (2.1) — the outbound-fetch guard on every caller-supplied ?url= (/art, /stream, /radio_stream) and every redirect hop. Refuses private / loopback / link-local destinations unless the host is a device already in SERVERS/RENDERERS (so the LocalFs file server on :8200 stays reachable) and refuses non-http(s) schemes. Before it, /stream relayed ANY internal HTTP service's body verbatim and /art's error text was an open/closed/filtered port oracle. Refusals are logged; the caller gets a uniform failure.                                                                                                                                                  |
| P/g35 | dlna_credits.py                                          | Songwriting credits (2026-09-20): is this composer/lyricist a NAME or machine junk? DISPLAY-only — the file tag and the tracks row are untouched. 8% of this library's credits are scene adverts (www.t.me/...). REJECTED rule, kept as a test: "credits with no A-Za-z letter are junk" matched 55 rows, every one a real composer (■■■■■■■■■■■, ■■■■■■■■■■).                                                                                                                                                                                                                                                                                                                    |
| P/g36 | dlna_mbid.py                                             | The artist → MusicBrainz-id decision, pure. Every external source keys off this id, so one wrong match is wrong in four places and never invites correction — it REFUSES rather than guesses. & is never split (band names contain it), commas never at all, every dash folds to a space (MB spells names with typographic dashes), aliases only when nothing matched by name, and two artists sharing a name are refused.                                                                                                                                                                                                                                                        |
| P/g37 | dlna_lineup.py                                           | "Who was in the band when this was recorded?" — pure date-intersection over membership. Exists because per-recording performer credits cover only ~17% of tracks while membership is near-universal. Members REJOIN (Richard Wright has two stints), and no year means no claim.                                                                                                                                                                                                                                                                                                                                                                                                  |
| P/g38 | dlna_artist_fetch.py                                     | The artist-metadata sources — MusicBrainz (identity, life-span, genres, members), Wikipedia (bio, stored only WITH its url: CC BY-SA attribution), Last.fm (best-known, needs an application key). Parsing is split from HTTP so the judgement is testable. Owns the `member of band` DIRECTION rule: backward = a group's members, forward = a person's bands.                                                                                                                                                                                                                                                                                                                   |

| Code         | File / location         | Responsibility                                                                                                                                                                                                                         |
|--------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P/g39</b> | dlna_asgi_artist.py     | GET /api/artist_info — the artist panel's ONE request. Resolves the line-up for the track's year and LABELS it (inferred vs credits) so the client cannot promote a guess to a fact. Own module: dlna_asgi_browse is at 397/400 lines. |
| <b>P/g40</b> | dlna_library_artists.py | ArtistsMixin — artist_meta (id + display facts) and artist_members (line-up). Survives clear(uhn): each row cost a rate-limited round-trip. Keyed by the NORMALISED name so spelling variants are one question, not three.             |

Maintenance tools (T/\*) — tools/\*.py

Operate directly on library.db and/or the music root. All default to dry-run / report-only; deletions move to Trash unless --hard-delete. Each has a unit-test sibling tools/test\_\*.py.

| Code  | File                          | Purpose & key options                                                                                                                                                                                                                                                                                                  |
|-------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T/a1  | regen_schema.py               | Regenerate the committed schema.sql artifact after any schema change. --check = fail if stale (the test_schema_sync gate).                                                                                                                                                                                             |
| T/a2  | prune_empty_music_dirs.py     | Trash directories whose subtree has zero music files. --dry-run -v --limit N --exts a,b --hard-delete -y.                                                                                                                                                                                                              |
| T/a3  | find_corrupt_audio.py         | Flag files with bad/zero magic bytes for their extension. Writes corrupt-audio.txt. --trash --hard-delete --limit --exts --out -y.                                                                                                                                                                                     |
| T/a4  | find_duplicate_audio.py       | Find duplicate recordings on disk (same acoustid metadata); ranks a winner by bit-depth/sample-rate/size. --trash --hard-delete -v -y.                                                                                                                                                                                 |
| T/a6  | relink_orphan_overrides.py    | Relink metadata_overrides orphaned by a UPnP rescan via d-id + fuzzy (artist,title). --apply --db.                                                                                                                                                                                                                     |
| T/a7  | relink_playlists_to_locals.py | Repoint playlist_tracks/favourites at RoHaLocalFS by normalised metadata; prune no-match. --apply --no-backup --no-prune-favs -y.                                                                                                                                                                                      |
| T/a8  | audit_override_mismatches.py  | Delete acoustid overrides whose artist AND title both mismatch the track (d-id collision repair). --clean --top -y.                                                                                                                                                                                                    |
| T/a9  | correct_year_drift.py         | Rewrite metadata_overrides.year to the earliest plausible year evidenced by another copy in the library. --apply --top --db -y.                                                                                                                                                                                        |
| T/a10 | improve_song_years.py         | Query MusicBrainz for each song's earliest recording year → song_year_cache → apply. --lookup --apply --limit -v.                                                                                                                                                                                                      |
| T/a11 | locals_scan.py                | Standalone LocalFs indexer (CLI). NOTE: do not run against the live DB without a base_url — leaves placeholder URLs.                                                                                                                                                                                                   |
| T/a12 | locals_serve.py               | Standalone LocalFs file-server launcher for testing the :8200 serving path.                                                                                                                                                                                                                                            |
| T/a13 | beets_enrich.py               | Run the beets tag-in-place enrichment batch (docs/enrichment.md). Safe wrapper around `beet import`: enforces write:yes/copy:no/move:no before any write. --write-config --quiet --timid --album --revisit --reindex --gateway --dry-run -y.                                                                           |
| T/a14 | post_beets_reindex.py         | The post-beets step: drop the historical AcoustID metadata_overrides (LocalFs URLs are path-stable, so old acoustid rows would re-mask beets' fresh tags) then reindex LocalFs. --apply --dry-run --no-clean --no-reindex --no-backup --udn --gateway --db -y.                                                         |
| T/a15 | artist_mbid.py                | Resolve every artist to a MusicBrainz id — the keystone the rest of the artist metadata hangs off. Tags first (free, exact), then name search. Live: 2,889 artists, 87.8% resolved, 85.9% search acceptance. Resumable (sticky notfound). --apply --limit N --tags-only -v.                                            |
| T/a16 | artist_meta.py                | Turn each id into the panel's facts: life-span, birthplace, genres, biography, best-known, and a band's members with instruments + dates. Live: 2,889 in 3h37m, 0 errors. SET LASTFM_API_KEY FIRST — meta_fetched_at marks an artist done, so a keyless pass leaves them all without best-known. --apply --limit N -v. |

LAN devices (D/\*), external services (E/\*), scheduled jobs (J/\*)

| Code | Name                                     | Role / protocol                                                                                                                                                                                                                   |
|------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D/1  | Naim Uniti                               | UPnP MediaRenderer. Receives AVTransport+RenderingControl SOAP (control/volume); pulls audio bytes over HTTP Range from :8200; also browses gateway playlists as a control point. Gateway is NOT in its audio path → bit-perfect. |
| D/2  | UPnP MediaServer                         | MinimServer / generic UPnP (AssetUPnP decommissioned). Browsed via P/g10 UpnpProvider + P/g13 ContentDirectory SOAP. Kept for any non-LocalFs source.                                                                             |
| D/3  | Media files                              | /Volumes/SAMDATA/Music + /Volumes/SAMDATA/GWMovies (external drive, read-only). Audio: indexed by P/g9, tags read by locals provider, bytes served by P/g11. Video: scanned by P/g31, served by P/g11 (/locals/video).            |
| D/4  | fpcalc · ffmpeg/ffprobe                  | Local CLI binaries (brew). fpcalc fingerprints for the beets tool (T/a13); ffmpeg/ffprobe probe metadata, extract posters and run the on-demand HLS transcode (P/g32). Internal/local — not network.                              |
| D/5  | LG TV (webOS)                            | DLNA control point + player. Browses the gateway's /gw MediaServer (incl. the Videos folder) and pulls bytes from :8200. Plays HEVC/MKV natively — the PWA transcode path is browser-only.                                        |
| E/x1 | musicbrainz.org                          | GET /ws/2/release-group — MBID + original year. UA + 1.1 s rate limit required.                                                                                                                                                   |
| E/x2 | coverartarchive.org                      | HEAD /release-group/{mbid}/front-500 — cover presence.                                                                                                                                                                            |
| E/x3 | Irclib.net                               | GET /api/get — on-demand lyrics for the playing track.                                                                                                                                                                            |
| E/x4 | *.api.radio-browser.info                 | GET /json/stations/search — internet-radio catalogue (HLS filtered).                                                                                                                                                              |
| E/x5 | api.acoustid.org                         | fingerprint → MusicBrainz metadata — used ONLY by the beets tool (T/a13) now; the in-process AcoustID worker was removed in 2.0.                                                                                                  |
| E/x6 | nominatim.openstreetmap.org              | GET /reverse — GPS → place name for video display titles (P/g33). UA + 1.1 s rate limit per OSM policy; sticky cache in geocode_cache.                                                                                            |
| E/x7 | en.wikipedia.org                         | GET /api/rest_v1/page/summary/{title} — artist biography for the artist panel. CC BY-SA: the bio is DROPPED if its url is missing, because the link is the attribution. Disambiguation stubs refused.                             |
| E/x8 | ws.audioscrobbler.com                    | Last.fm artist.getTopTracks — “best known for”, ranked by real listening. Needs an APPLICATION key (LASTFM_API_KEY); an account password is NOT an API credential. Absent → the block is omitted.                                 |
| J/1  | com.roha.dlna-gateway                    | LaunchAgent that runs the gateway. Restart: launchctl kickstart -k gui/\${id -u}/com.roha.dlna-gateway.                                                                                                                           |
| J/2  | com.roha.dlna-cert-renew + renew-cert.sh | Weekly TLS cert renewal (Mon 04:30; no-op unless <30 days). cert-renewal.log.                                                                                                                                                     |
| J/4  | setup.sh                                 | venv setup + run / restart / probe. --run --restart --no-browser --debug --probe URL --list-devices --reset-devices. See the setup.sh options reference.                                                                          |

Commands & options used in the drawing

| Context            | Command                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Run (J/4)          | ./setup.sh --run [--no-browser] [--debug] [--probe http://...] [--list-devices] [--reset-devices]                                                  |
| Restart (J/4)      | ./setup.sh --restart (refresh venv/deps + launchctl kickstart -k gui/(id -u)/com.roha.dlna-gateway)                                                |
| Cert (J/2)         | ./renew-cert.sh [--force] · launchctl kickstart gui/(id -u)/com.roha.dlna-cert-renew                                                               |
| Subsonic auth      | launchctl setenv SUBSONIC_PASSWORD ... ; launchctl getenv SUBSONIC_PASSWORD                                                                        |
| Full test suite    | python tests/run_all.py [--offline   --frontend   --frontend-only   http://host:8765]                                                              |
| Unit / Playwright  | python3 -m unittest tests.test_player ... ; .venv/bin/pytest tests/frontend -v                                                                     |
| Chaos (live)       | python3 tests/chaos.py --iterations N --workers M [--seed S] [--quiet] [--base https://host:8443]                                                  |
| Load (live)        | python3 tests/load_stream.py --concurrency 40 --count 80 [--gateway https://127.0.0.1:8443 --insecure] [--max-p95 6] (threadpool-starvation guard) |
| Safari / iOS smoke | .venv/bin/python tests/frontend/safari_smoke.py · ios_sim_smoke.py (Appium :4723 + a booted Simulator) — opt-in, not in run_all.py                 |
| Schema gate (T/a1) | python3 tools/regen_schema.py [--check]                                                                                                            |
| Module self-tests  | python dlna_discovery.py   dlna_content.py   dlna_library.py   db_pool.py   dlna_player.py                                                         |
| Range / 206 check  | curl -r 0-1023 -D - http://<host>:8200/locals/stream/<id> -o /dev/null                                                                             |
| Tools (T/a2–a12)   | python3 tools/<tool>.py [--dry-run --apply --trash --hard-delete --limit N --exts a,b -v -y] (see the tools table above)                           |

Ports (2.0): 8443 HTTPS — Hypercorn TLS + HTTP/2 (ALPN), tailscale cert · 8765 plain HTTP, incl. the Naim/LG-facing /gw/\* UPnP surface (Cleanup C folded the old :8770 device tier into this bind; SSDP advert points here) · 8200 RoHaLocalFS file server (audio + /locals/video) · 26125 (legacy AssetUPnP, decommissioned). The HTTP/2 + app-owned-TLS roadmap is now DONE — Hypercorn terminates TLS/h2 natively.

## Tool options reference — what each flag does

Every option for every maintenance tool, with its meaning. Codes match the diagram (T/aN) and the tools table. Unless noted, file-deleting tools default to dry-run / report-only and move to the macOS Trash (recoverable ~30 days) — **--hard-delete** is the permanent, non-recoverable variant.

### T/a1 regen\_schema.py — Regenerate the committed schema.sql.

| Option    | Meaning                                                                        |
|-----------|--------------------------------------------------------------------------------|
| (no args) | Rewrite schema.sql from the live LibraryDB schema.                             |
| --check   | Don't write; exit non-zero if schema.sql is stale (the test_schema_sync gate). |

### T/a2 prune\_empty\_music\_dirs.py — Trash directories whose subtree has zero music files.

| Option         | Meaning                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------|
|                | Music root to walk (positional).                                                                         |
| --dry-run      | Print decisions; touch nothing.                                                                          |
| -v / --verbose | Also log every KEPT directory (default: only deletions print).                                           |
| --limit N      | Stop after evaluating N dirs; when the limit is hit, NO deletions run (a partial picture = safety belt). |
| --exts a,b,c   | Override the music-extension set (commas, dot optional).                                                 |
| --hard-delete  | Permanent rm -rf instead of Trash. NOT recoverable.                                                      |
| -y / --yes     | Skip the confirmation prompt.                                                                            |

### T/a3 find\_corrupt\_audio.py — Flag files whose magic bytes are wrong/zero for their extension.

| Option         | Meaning                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------|
|                | Music root to scan (positional).                                                                 |
| -v / --verbose | Also log every OK file (default: only corrupt).                                                  |
| --limit N      | Stop after scanning N files (0 = no limit); halts the delete step when hit.                      |
| --exts a,b,c   | Override the audio-extension list.                                                               |
| --out PATH     | Where to write the corrupt-paths list (default ./corrupt-audio.txt; pass /dev/null to suppress). |
| --trash        | Move flagged files to the Trash.                                                                 |
| --hard-delete  | Permanent unlink instead of Trash. Mutually exclusive with --trash.                              |
| -y / --yes     | Skip the confirmation prompt before deleting.                                                    |

### T/a4 find\_duplicate\_audio.py — Find duplicate recordings on disk (same acoustid metadata); keeps a ranked winner.

| Option         | Meaning                                          |
|----------------|--------------------------------------------------|
|                | Music root to scan (positional).                 |
| -v / --verbose | Per-URL ambiguity / not-found logs.              |
| --trash        | Move the loser files to the Trash (winner kept). |
| --hard-delete  | Permanent delete of losers. NOT recoverable.     |
| -y / --yes     | Skip the confirmation prompt.                    |

### T/a6 relink\_orphan\_overrides.py — Relink metadata overrides orphaned by a UPnP rescan (d-id + fuzzy artist/title).

| Option    | Meaning                        |
|-----------|--------------------------------|
| (default) | Dry-run preview — no mutation. |
| --apply   | Perform the relinks.           |
| --db PATH | library.db path.               |

### T/a7 relink\_playlists\_to\_locals.py — Repoint playlists/favourites at RoHaLocalFS by normalised metadata.

| Option          | Meaning                                                       |
|-----------------|---------------------------------------------------------------|
| (default)       | Dry-run preview.                                              |
| --apply         | Commit the relinks (auto-backs up library.db first).          |
| --no-backup     | Skip the automatic library.db backup on --apply.              |
| --no-prune-favs | Keep album_favourites that no longer match any LocalFs album. |
| -y / --yes      | Skip the confirmation prompt.                                 |

### T/a8 audit\_override\_mismatches.py — Delete acoustid overrides whose artist AND title both mismatch the track.

| Option     | Meaning                                    |
|------------|--------------------------------------------|
| (default)  | Dry-run, lists the top 30 suspects.        |
| --clean    | Delete the suspect acoustid override rows. |
| --top N    | Show N rows (0 = full list).               |
| -y / --yes | Non-interactive (skip the confirm).        |

T/a9 correct\_year\_drift.py — Rewrite metadata\_overrides.year to the earliest plausible year from another copy in the library.

| Option     | Meaning                                     |
|------------|---------------------------------------------|
| (default)  | Dry-run, top 30 candidates.                 |
| --apply    | Write the corrections (as source='manual'). |
| --top N    | Preview N candidates (0 = all).             |
| --db PATH  | library.db path.                            |
| -y / --yes | Non-interactive apply.                      |

T/a10 improve\_song\_years.py — Query MusicBrainz for each song's earliest recording year → cache → apply.

| Option         | Meaning                                                                     |
|----------------|-----------------------------------------------------------------------------|
| (default)      | Dry-run preview — no MB calls, no DB writes.                                |
| --lookup       | Query MB for uncached (artist,title) groups; fill song_year_cache (sticky). |
| --apply        | Write cached hits onto tracks (metadata_overrides.year, source='manual').   |
| --limit N      | Cap the number of groups queried (for testing).                             |
| -v / --verbose | Per-group logging.                                                          |

T/a11 localfs\_scan.py — Standalone LocalFs indexer (CLI). Don't run against the live DB without a base\_url — it leaves placeholder URLs.

| Option         | Meaning                                                                         |
|----------------|---------------------------------------------------------------------------------|
| --root PATH    | Music root to scan (default \$LOCALFS_MUSIC_ROOT or /Volumes/SAMDATA/Music).    |
| --force        | Ignore the (mtime, size) cache and re-tag every file. Use after schema changes. |
| --compare      | Skip scanning — just print tracks/albums per UDN currently in library.db.       |
| --db PATH      | library.db path.                                                                |
| -v / --verbose | Per-file logging.                                                               |

T/a12 localfs\_serve.py — Standalone RoHaLocalFS file-server launcher (test the :8200 path).

| Option         | Meaning                                                           |
|----------------|-------------------------------------------------------------------|
| --port N       | HTTP port (default 8200 / \$LOCALFS_PORT).                        |
| --host ADDR    | Listen address (default 0.0.0.0 so the Naim can reach it).        |
| --db PATH      | library.db path.                                                  |
| --root PATH    | Allowed music root; repeatable. Defaults to \$LOCALFS_MUSIC_ROOT. |
| -v / --verbose | Request logging.                                                  |

T/a13 beets\_enrich.py — Run the beets tag-in-place enrichment batch (docs/enrichment.md). Safe wrapper around `beet import`. Deps: brew install chromaprint beets (NOT pip — a Homebrew python upgrade wipes a pip install; that is how the 2026-06 install died), then add musicbrainzngs + pyacoustid to the keg venv (beets 2.x pluginized MusicBrainz — the musicbrainz plugin + musicbrainzngs are REQUIRED or it matches nothing).

| Option               | Meaning                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (no --quiet/--timid) | Interactive: beets prompts you per album (apply / skip / ...); strong matches auto-apply.                                                                |
| --write-config       | Write the prog-tuned tag-in-place ~/.config/beets/config.yaml (enables the musicbrainz metadata plugin; backs up any existing) and exit. Run this first. |
| --music-root PATH    | Library root to import (default /Volumes/SAMDATA/Music).                                                                                                 |
| --album PATH         | Import a single album directory instead of the whole root.                                                                                               |
| --config PATH        | beets config path (default ~/.config/beets/config.yaml).                                                                                                 |
| --quiet              | Unattended BULK pass: auto-accept only strong matches (≥ strong_rec_thresh 0.80), skip the rest, no prompts (beet import -q).                            |
| --timid              | Most cautious: prompt for EVERY match (per-change review). Mutually exclusive with --quiet.                                                              |
| --revisit            | Re-import a directory beets already recorded as done (-I / noincremental). Use after re-tagging an album.                                                |
| --reindex            | After import, find the LocalFs server UDN via /api/servers and POST /api/index/rebuild so the gateway picks up the new tags.                             |
| --gateway URL        | Gateway base URL for --reindex (default http://127.0.0.1:8765).                                                                                          |
| --udn UDN            | Explicit server UDN to reindex (default: auto-pick the uuid:localfs-* server).                                                                           |
| -n / --dry-run       | Print the resolved beet command + safety report; do NOT invoke beets (beets has no true dry-run of its own).                                             |
| -y / --yes           | Skip the in-place-write backup-warning confirmation.                                                                                                     |

T/a14 post\_beets\_reindex.py — After beets has tagged files in place, make its work visible: clear the AcoustID metadata\_overrides (LocalFs URLs are PATH-stable, so the COALESCE pass would otherwise re-lay old acoustid rows over beets' fresh tags) THEN reindex LocalFs. source='manual' is never touched; notfound/video\_skip carry NULL metadata so they mask nothing.

| Option         | Meaning                                                                                  |
|----------------|------------------------------------------------------------------------------------------|
| (default)      | Dry-run: print the override breakdown + the planned clear/reindex; change nothing.       |
| --apply        | Actually delete acoustid overrides + start the reindex (auto-backs up library.db first). |
| -n / --dry-run | Explicit preview alias (the default when --apply is absent).                             |
| --no-clean     | Skip clearing overrides (reindex only).                                                  |
| --no-reindex   | Skip the reindex (clean only).                                                           |
| --no-backup    | Skip the library.db backup before deleting.                                              |
| --udn UDN      | Server UDN to reindex (default: auto-pick uuid:localfs-*).                               |
| --gateway URL  | Gateway base URL (default http://127.0.0.1:8765).                                        |
| --db PATH      | library.db path.                                                                         |
| -y / --yes     | Skip the confirmation prompt.                                                            |

J/4 setup.sh — Bootstrap + run the gateway. Finds Python 3.14+, creates/repairs the .venv, installs requirements.txt, then either runs the gateway (--run, exec in the foreground), restarts the launchd-managed copy (--restart), or just finishes setup. Unknown flags after --run are forwarded verbatim to dlina\_gateway.py.

| Option             | Meaning                                                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (no flags)         | Set up only: venv + dependencies, then print the 'how to run' summary. Does not start anything.                                                                                                                                                                                                                            |
| --run              | Set up, then exec the gateway in the foreground on :8765 (blocks; Ctrl-C to stop). All flags below it are passed through to dlina_gateway.py.                                                                                                                                                                              |
| --restart          | Refresh the venv/deps, then restart the launchd gateway via launchctl kickstart -k gui/\$(id -u)/com.roha.dlina-gateway (the launchd-correct restart — a bare kill races launchd's respawn). Aborts with install hints if the LaunchAgent isn't loaded. Mutually informative with --run; --restart wins if both are given. |
| --no-browser       | (forwarded) Don't auto-open the browser on start.                                                                                                                                                                                                                                                                          |
| --debug            | (forwarded) Verbose logging.                                                                                                                                                                                                                                                                                               |
| --probe http://... | (forwarded) Add a UPnP server manually by its device-description URL.                                                                                                                                                                                                                                                      |
| --list-devices     | (forwarded) Print the known-devices table and exit.                                                                                                                                                                                                                                                                        |
| --reset-devices    | (forwarded) Wipe the device DB and exit.                                                                                                                                                                                                                                                                                   |
| --port N           | (forwarded) HTTP port (default 8765).                                                                                                                                                                                                                                                                                      |